In America, when we turn sixteen, we're finally given the "freedom" to be autonomous, leave the house and get around town without hitching a ride from our parents (given how car dependent America is compared to other European countries, car infrastructure has actually stripped us of our freedom in many ways but that's a different conversation). What grants us this freedom are cars, vehicles meant to get us from point A to point B. However, overtime, manufacturers find the need to add "quality of life" improvements and other technological additions to an automobile that is simply meant to move us around. This complicates an already somewhat complicated piece of technology. Cars now have all kinds of sensors recording the driver and passenger. Not only are they collecting private information, but cars now depend on updates and subscriptions to operate. In the interest of advancing the use of information systems in cars, cars now collect an unnecessary amount of data, and are reliant on technologies that hinder the experience of the rider.

Cars should be simple machines, but now they are essentially computers on wheels. Like many computers, they are collecting a great deal of personal data of their drivers and passengers. Mozilla recently conducted an in depth privacy report. According to them, all 25 car brands they reviewed collect personal data. The data collected pertains to how drivers interact with their car including apps and connected services, how fast drivers go, where drivers go, and even their sex life (or maybe lack thereof). The data is used to create inferences on their customers such as interests and knowledge. Most of the car brands also have the ability to share or sell driver data. Most of the time drivers don't have control over their personal data. Not only is this data collection a violation of our privacy, but of course this unnecessary data collection is open to malicious actors if they are determined. Unfortunately, there's not much consumers can do to combat this. If someone already has a privacy invasive car, at best they can limit their use of the car's apps and possibly turn off certain permissions. In terms of privacy, it would be ideal to depend on a less technologically advanced vehicle. While someone's 2008 Honda Accord may not be flashiest or fastest car, at least it isn't spying on them. Another alternative could be getting around by ebike, which is also much more economical than a combustion vehicle.

Not only are these computers on wheels spying on everyone, but just like laptops and desktops, they won't work without subscriptions and updates. An electric car known as the Ford Mustang Mach-E recently went viral, someone on Reddit posted a picture of a failed update. Since the update was unsuccessful, the car was

unable to be driven in any capacity and the driver is asked to contact customer support (from there the car must be towed to the dealership). A car that costs over $60,000 is incapable of reverting to a previous update if it fails. To make matters worse, cars are also adopting predatory pricing schemes.

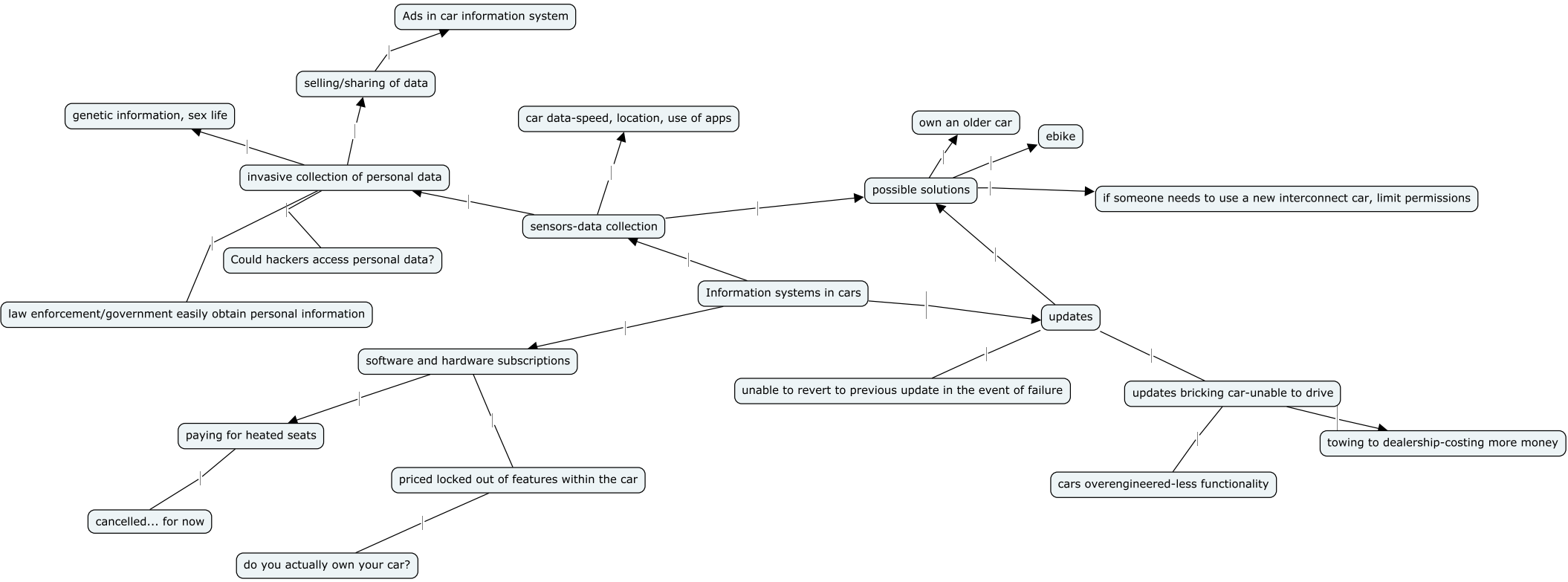
BMW was looking at the possibility of charging a subscription for heated seats (Hawkins *BMW drops plan to charge a monthly fee for heated seats*). Fortunately they decided not to go forward with this decision due to public outcry, but they said they will seek software based subscriptions. Even though they have not gone forward with this decision, Pandora's box has been opened and it's always possible for this to happen in the future with any car manufacturer. Financial software locks already exist in cars. The new Tesla Model S and X are offered in standard range and long range versions. Both versions come equipped with the same battery pack, yet consumers must pay more for to unlock the rest of the range. Of course it's generally normal to pay less for a car with less functionally, but often that car has those extra features completely stripped out.

As time progresses, technology should be innovated. In many ways, cars have come a long way. More efficient engines, the conversion from electric to combustion, and simple things such as a backup camera. Unfortunately, in many other ways cars have regressed and less functionality. Cars have become unnecessarily dependent on other forms of technology, hindering the ability to even drive the car in the event of a failed update. The technological dependence has also allowed manufactures to employ unfair pricing tactics to lock out features that come with the car. As of now, cars have taken away our freedom. We are moving towards a future in which we own less and have worse control over what we do own. When someone purchases a product, they should be able to control, repair, or modify as they so choose.

This was my first time using a mind map to articulate my ideas. I found that it helped me visual the processes of the information systems with a car. For instance I would connect the failures of car updates and the personal data collected with cars. On top of visualizing the processes, I reflected on the "slippery slope" of the processes, the consequences of these interconnected services within automobiles.

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Ads in car information system
selling/sharing of data
genetic information, sex life
invasive collection of personal data
car data-speed, location, use of apps
own an older car
ebike
possible solutions
if someone needs to use a new interconnect car, limit permissions
sensors-data collection
Could hackers access personal data?
law enforcement/government easily obtain personal information
Information systems in cars
updates
software and hardware subscriptions
unable to revert to previous update in the event of failure
updates bricking car-unable to drive
towing to dealership-costing more money
paying for heated seats
priced locked out of features within the car
cars overengineered-less functionality
cancelled... for now
do you actually own your car?